Transients through the 12 V front end: what you can and can't see

Bench test 2026-11-05: Arduino Giga R1 (firmware Giga_RemoteController_WiFi, dual-ADC build) reading the 12 V op-amp buffer board, preset OpAmpBuffer_12vboard (input = 7.005 x (A2 - A3)). Transients came from a Rigol DG852 Pro into the board input; the Giga was on USB to the PC. Every "recorded" trace in this document is real data from the board; 128 captures in total.

The short version

- **The front end is a low-pass filter with a 13.6 us time constant** (-3 dB at 10.9 kHz). Every transient comes out smoothed by it: a step takes 34 us to rise 10-90 %.
- **Pulses longer than about 50 us** are recorded with their true height and a duration you can read off the plot (the edges are rounded by the 34 us rise).
- **Pulses shorter than about 15 us lose height in proportion to their length**: what you see is their *area*. Recorded peak = height x width / 13.6 us. A 1 V, 1 us pulse reads about 62 mV, and it always looks about 12 us wide, however short it really was.
- **Noise sets the floor.** At 500 kS/s raw the noise is 33 mV rms; with SMOOTH 8 it is 11.7 mV. Single-shot, a 100 us pulse needs about 57 mV to stand out clearly (5 sigma) with SMOOTH 8, a 10 us pulse about 152 mV and a 1 us pulse about 1.4 V.
- **For weak transients use 500 kS/s with SMOOTH 8.** It sees everything raw 500 kS/s sees, at 2.8x lower noise. Lower sample rates without SMOOTH can step right over a short event.
- **Repeating transients can be averaged**: averaging N events lowers the floor by sqrt(N); 16 events take a pulse that is invisible in one shot to clearly visible.
- **Fast ringing shrinks with frequency**: a 10 kHz ring keeps about three quarters of its size, 30 kHz about a third, 100 kHz about a tenth (still visible raw) and 300 kHz only a few % (at the noise floor). SMOOTH 8 can't follow ringing above ~20 kHz; use raw 500 kS/s for that.

What a transient of a given length looks like

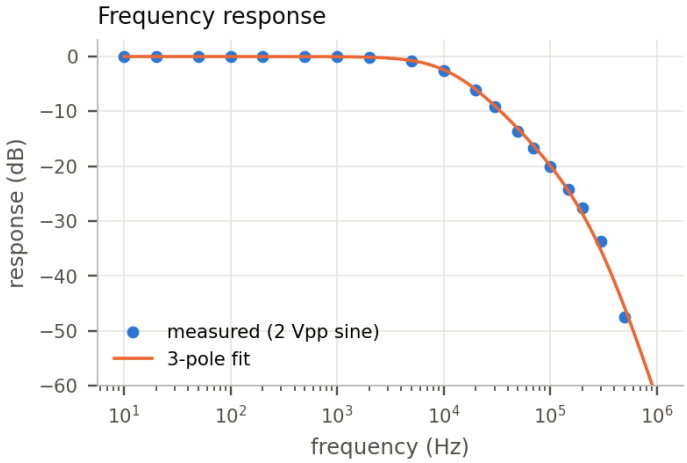
true duration	what the recording shows	smallest height seen single-shot (5 sigma) raw / SMOOTH 8
1 ms and longer	correct height, correct length, edges rounded (34 us rise)	160 mV / 57 mV
100 us	correct height, length readable, edges clearly rounded	160 mV / 57 mV
10 us	a rounded bump about 14 us wide (half height) at ~51 % of the true height	322 mV / 152 mV
1 us	a small bump about 12 us wide at ~6 % of the true height; only its area is real	2.6 V / 1.4 V

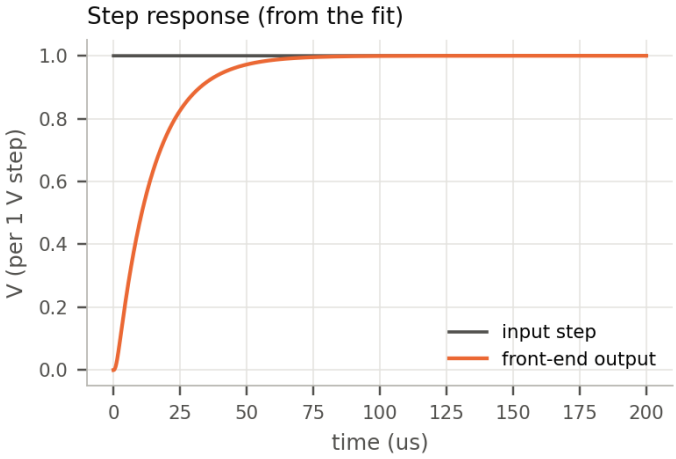
Smallest heights: 5 x the measured noise sigma at that setting, divided by the recorded fraction of the height for that duration (from the response, which the captures confirm within a few %). 5 sigma rarely happens by chance in a 10000-point record; a 3-sigma bump is visible when you know where to look. All heights are input-referred (at the board input).

How the test works and how to read the plots

The generator repeats each transient at a fixed interval (2 ms for most pulses) and the board takes an ordinary free-running capture. Because the generator timing is exact, every sample can be placed on the transient's own time axis: the capture is folded on the repeat interval and lined up with the expected response by a matched filter that uses every event in the record. The single-event traces are the first complete event in each record, not a chosen one.

- **Grey, top strip:** the true input, as the generator was set (edges 3 ns).
- **Blue dots:** what the board recorded for one event, input-referred, baseline removed.
- **Orange dashes:** what the measured frequency response predicts the board should record, including SMOOTH averaging. It is a prediction, not a fit: where blue sits on orange, the front end behaves as measured.
- **Grey band:** +/-3 sigma of the recording noise at that setting. A bump inside it can't be told from noise.



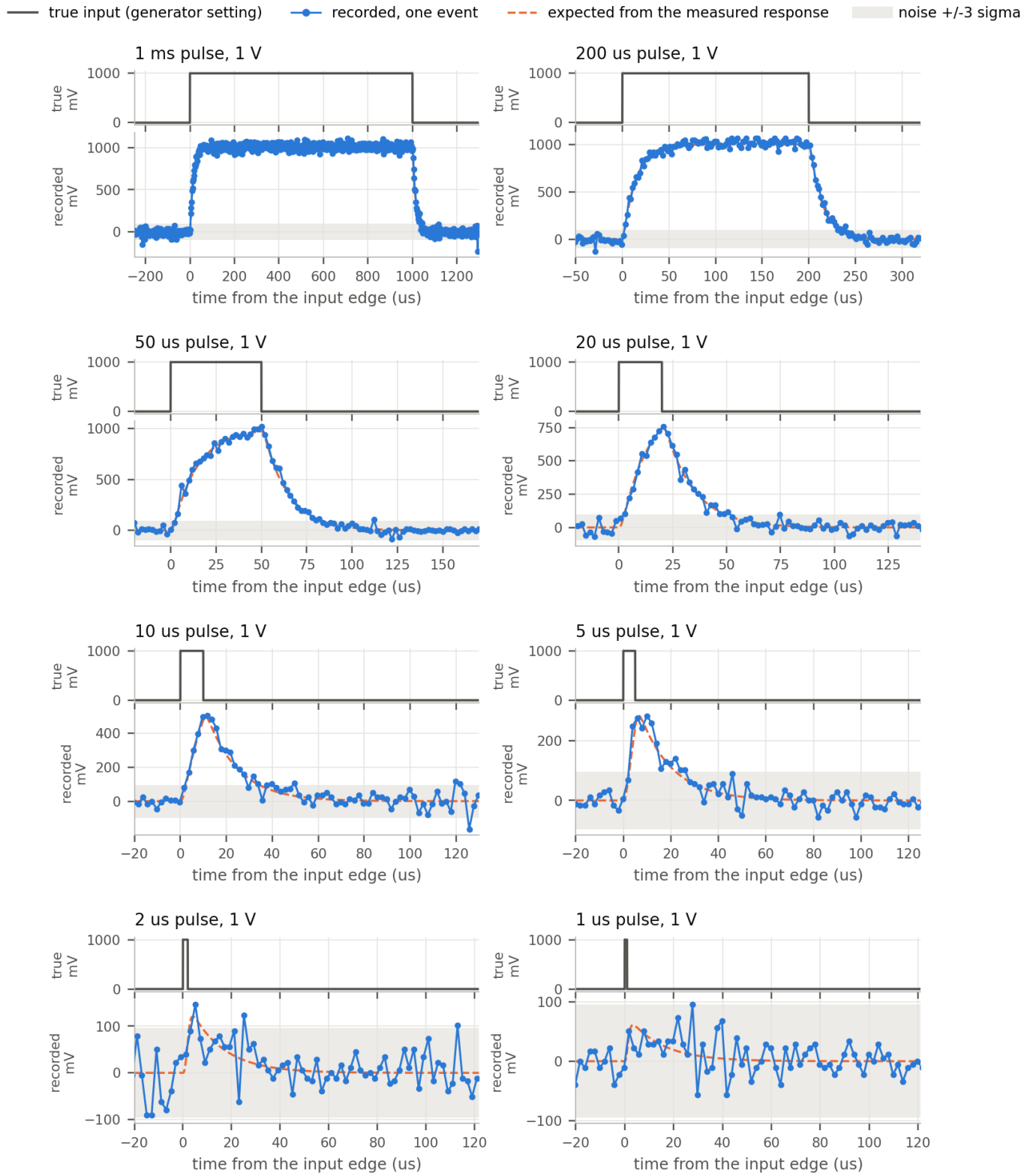


The response fit uses three real poles, at 11.7 kHz, 267 kHz and 267 kHz, matched to the sine sweep in the bench run (10 Hz to 500 kHz). The first pole does almost everything; the other two only matter above ~150 kHz. On the pulse captures the recorded amplitude matches the prediction within -0 to +5 % (5th to 95th percentile of the 28 captures with a clear signal, median +2.0 %), so the response can be used to predict transients that weren't tested.

setting	time per point	noise (input-referred rms)	notes
500 kS/s raw	2 us	32.6 mV	finest time detail
500 kS/s SMOOTH 8	16 us	11.7 mV	best all-round for weak transients
500 kS/s SMOOTH 16	32 us	8.4 mV	
100 kS/s raw	10 us	33.1 mV	
31.25 kS/s raw	32 us	32.5 mV	
10 kS/s raw	100 us	32.5 mV	

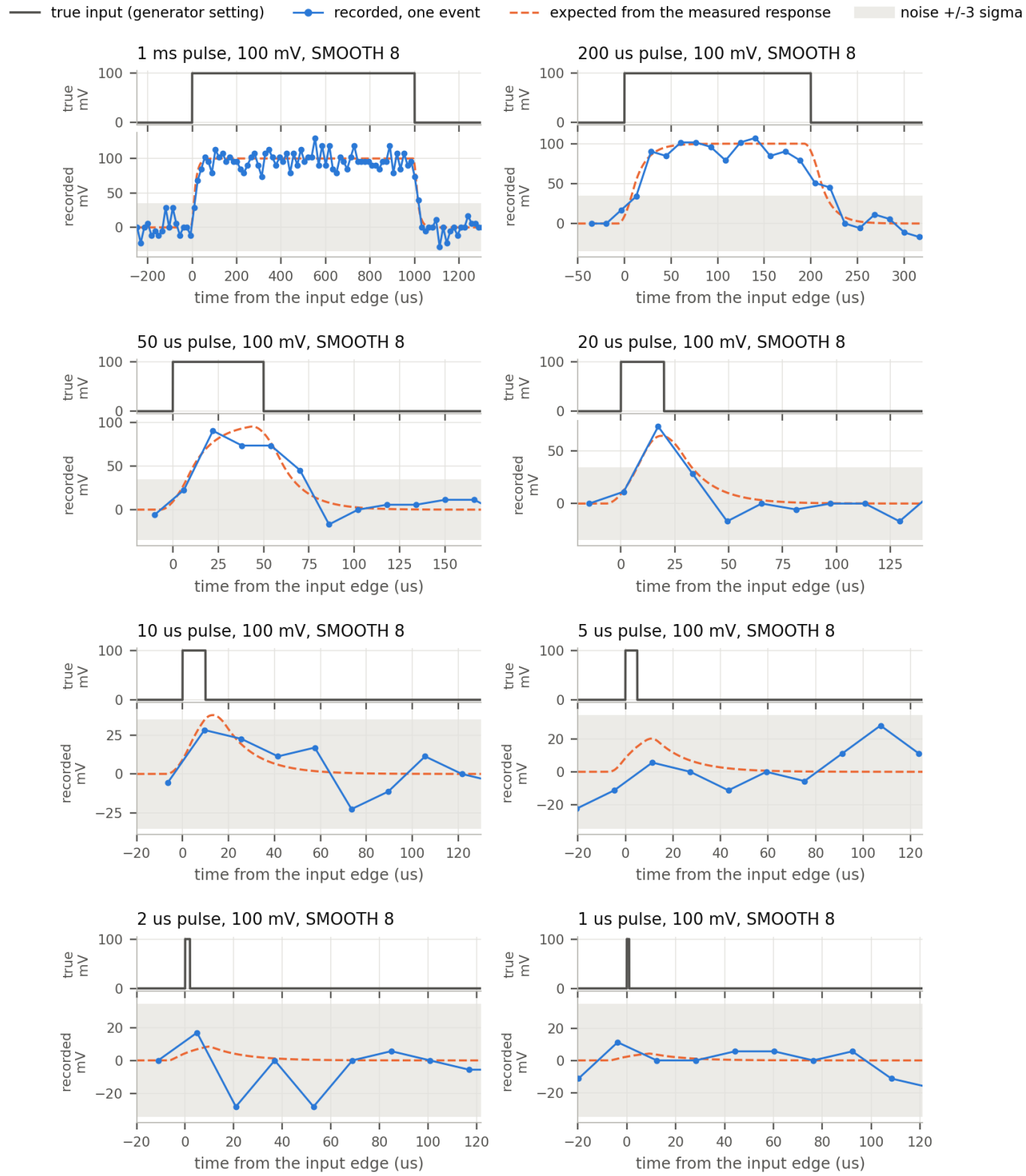
1. Pulse length, at 1 V, 500 kS/s raw

The same 1 V pulse from 1 ms down to 1 us. Long pulses come through with their height and length; below about 15 us the recorded bump stops getting narrower and just gets smaller. Note the different scales: the true strip is always 1 V tall.



2. Weak pulses: 100 mV, 500 kS/s SMOOTH 8

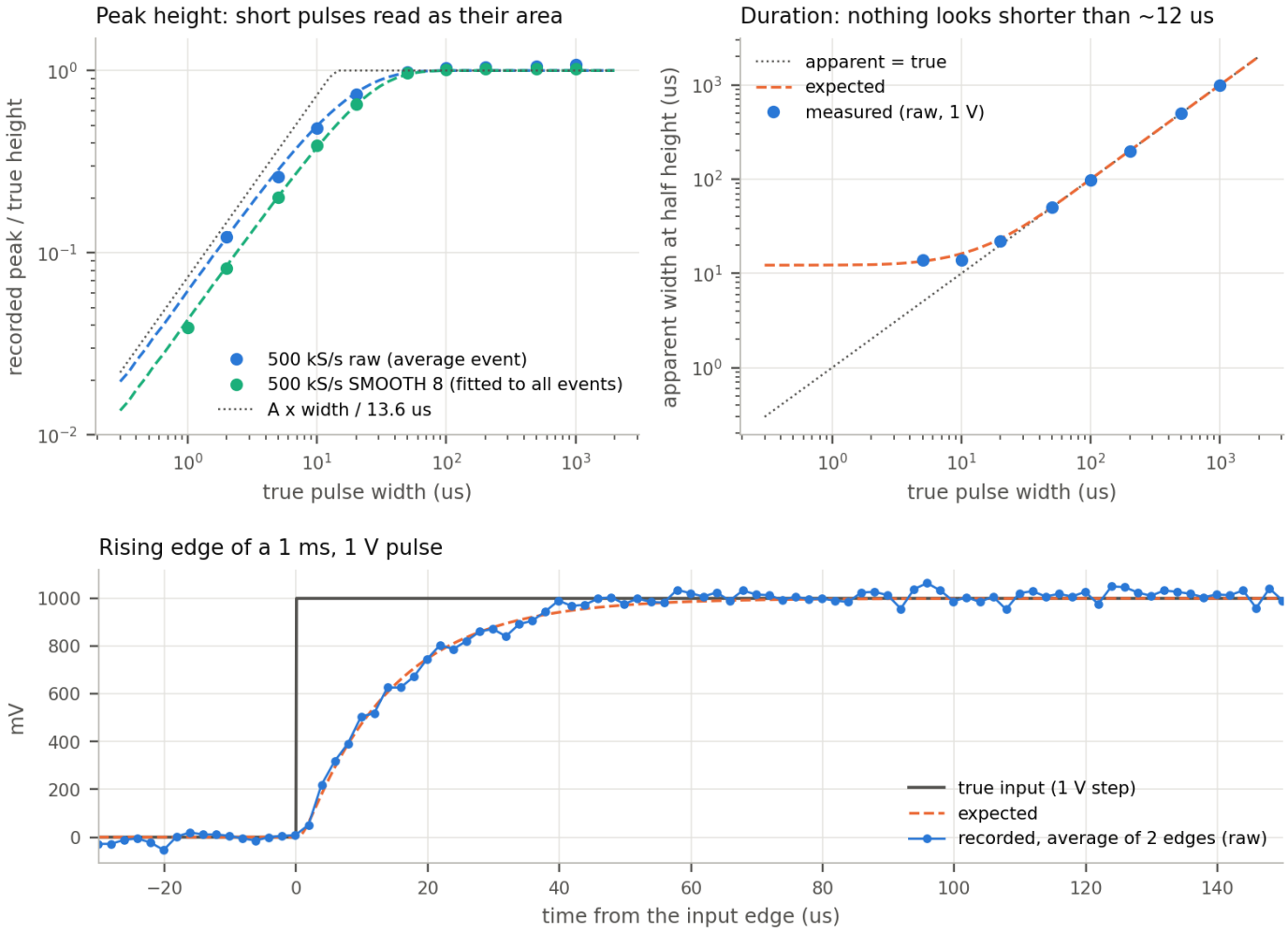
The same lengths at 100 mV with SMOOTH 8 (noise 11.7 mV). Down to 20 us the pulse is plain (5 sigma); 10 us is a bump at the edge of the noise (3 sigma); 5 us and shorter are lost in a single shot. SMOOTH 8 puts the points 16 us apart, so short events become two or three points: fine for seeing that something happened and how big its area was, not for its shape.



3. Short pulses are recorded as their area

Left: recorded peak as a fraction of the true height (measured from the average event of each 1 V record, dots; prediction, dashes). Above ~50 us the full height is reached. Below ~15 us the peak falls in direct proportion to width (dotted line: height x width / 13.6 us), so a 1 us, 1 V pulse and a 10 us, 100 mV pulse look the same. SMOOTH 8 lowers short peaks further (it averages over 16 us), but its noise is lower by more, so it still sees them better.

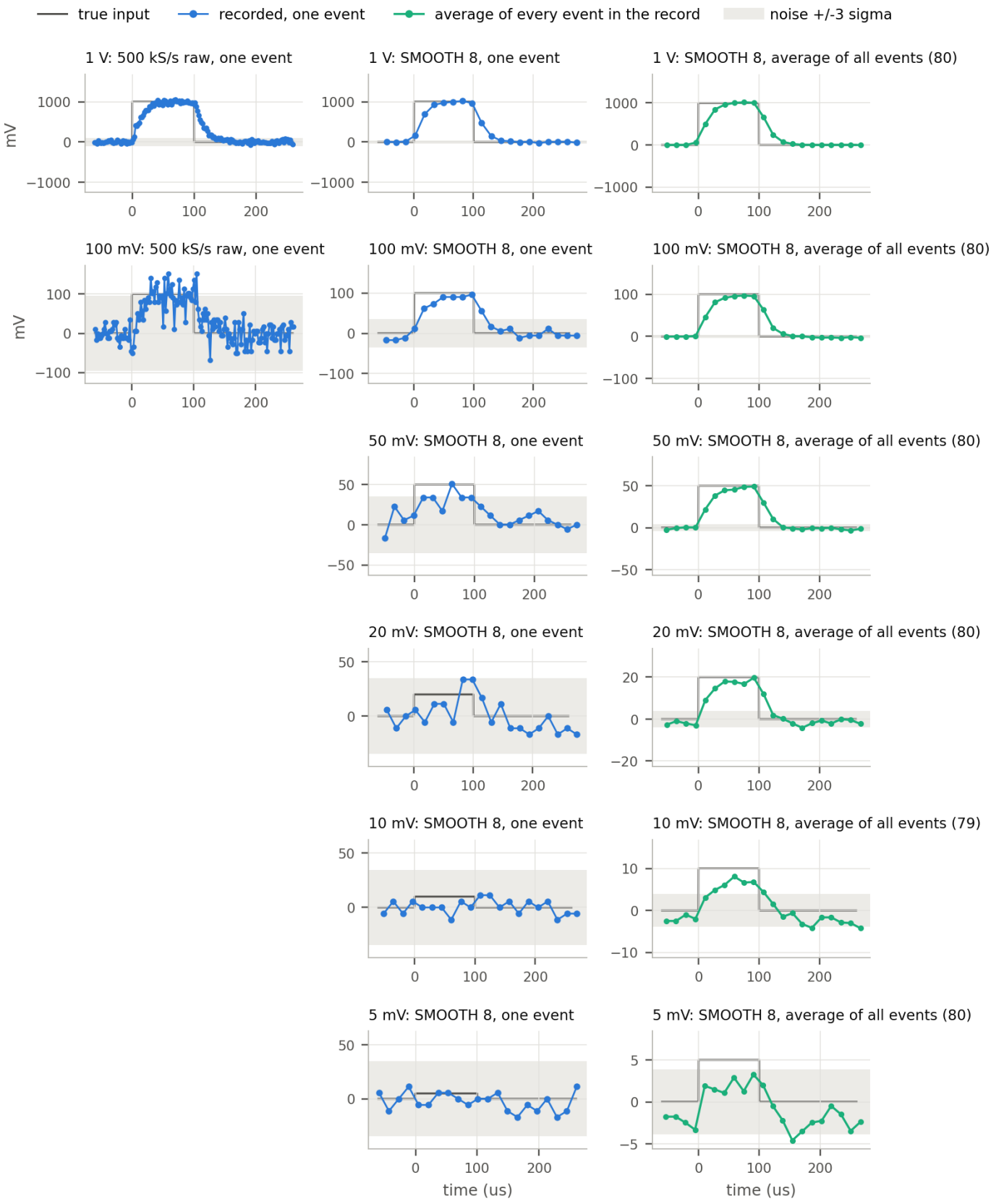
Right: the apparent duration (width at half height, raw, where the average event is clear). Nothing reads shorter than about 12 us; longer pulses read their true length plus a little. So an event that looks 12-15 us wide may really have been anything from 10 us down to a spike.



A step rises 10-90 % in 34 us measured (30 us predicted), so the start time of an event can be placed to a few us, but its edges always look rounded.

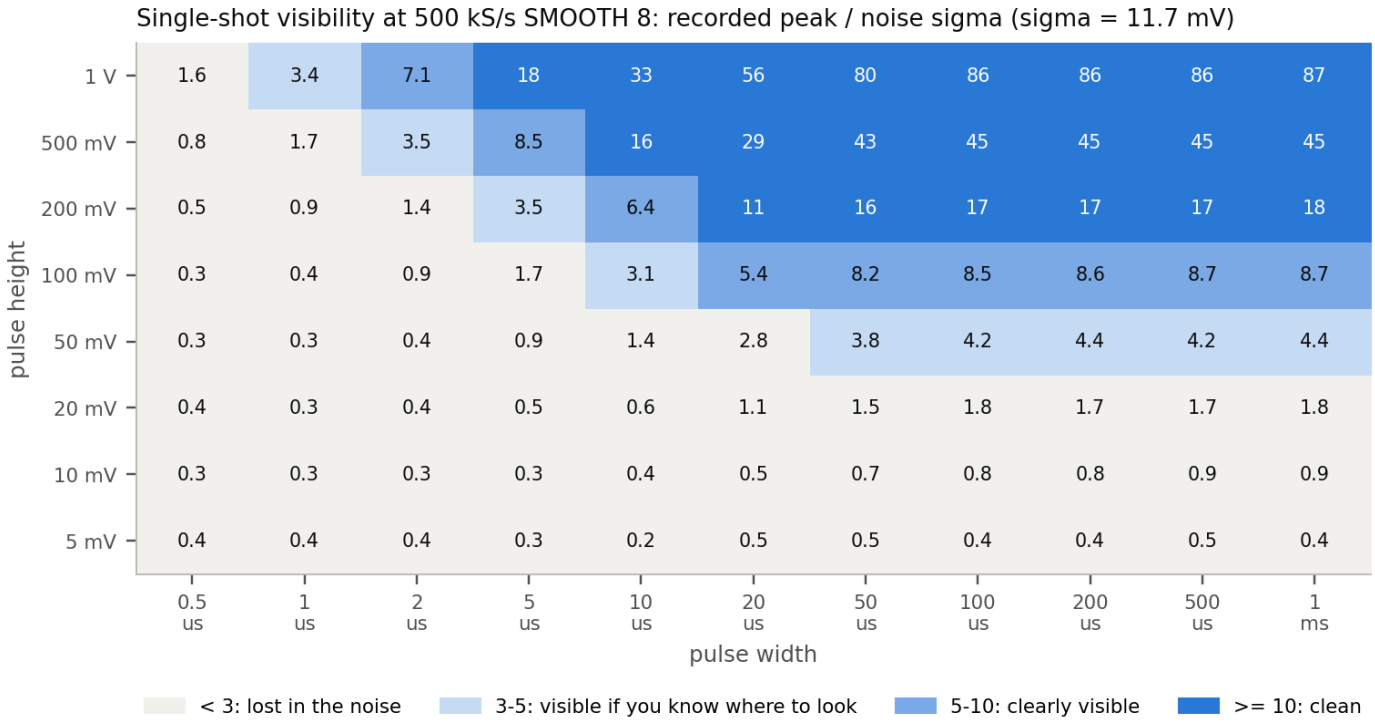
4. How weak can a transient be? (100 us pulse)

A 100 us pulse is long enough to be recorded at full height, so this isolates the noise. Raw 500 kS/s (left, noise 33 mV rms) loses it below about 100 mV. SMOOTH 8 (middle, 11.7 mV) still shows 50 mV, while 20 mV is at the edge of the noise. If the event repeats, averaging the events (right) shows 10 mV clearly and 5 mV just about. Raw was only captured at 100 mV and 1 V.

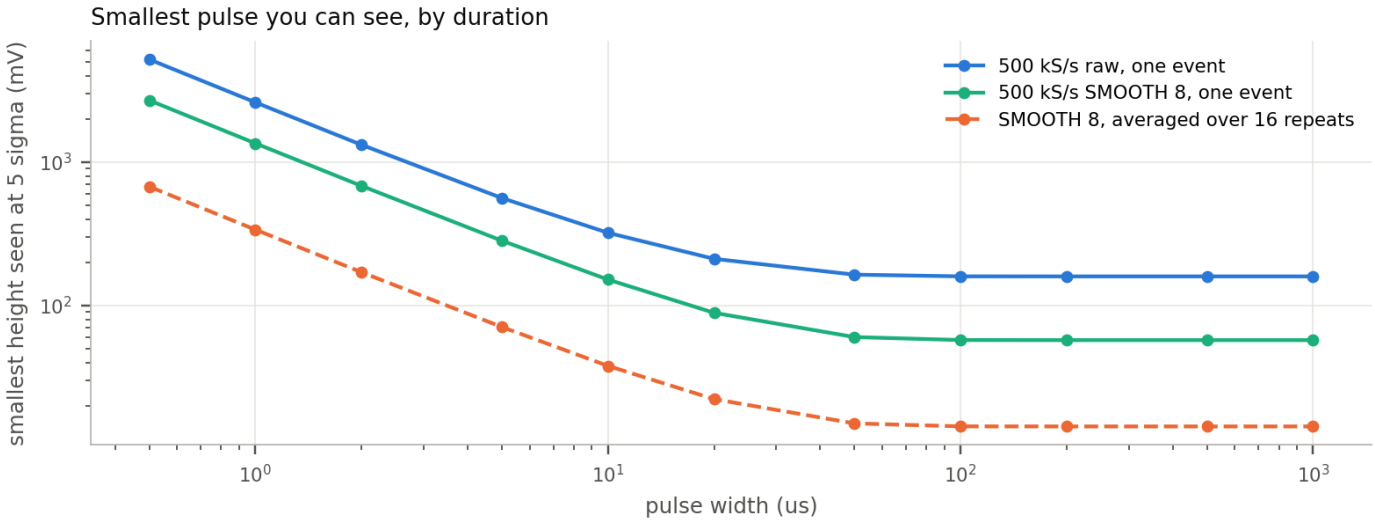

true input
recorded, one event
average of every event in the record
noise +/-3 sigma
1 V: 500 kS/s raw, one event
1 V: SMOOTH 8, one event
1 V: SMOOTH 8, average of all events (80)
100 mV: 500 kS/s raw, one event
100 mV: SMOOTH 8, one event
100 mV: SMOOTH 8, average of all events (80)
50 mV: SMOOTH 8, one event
50 mV: SMOOTH 8, average of all events (80)
20 mV: SMOOTH 8, one event
20 mV: SMOOTH 8, average of all events (80)
10 mV: SMOOTH 8, one event
10 mV: SMOOTH 8, average of all events (79)
5 mV: SMOOTH 8, one event
5 mV: SMOOTH 8, average of all events (80)
mV
time (us)

5. Visibility map: every length and height tested

Each cell is one capture at 500 kS/s SMOOTH 8: the recorded peak divided by the noise sigma, for a single event. Read it as: pick your transient's length (column) and size (row).

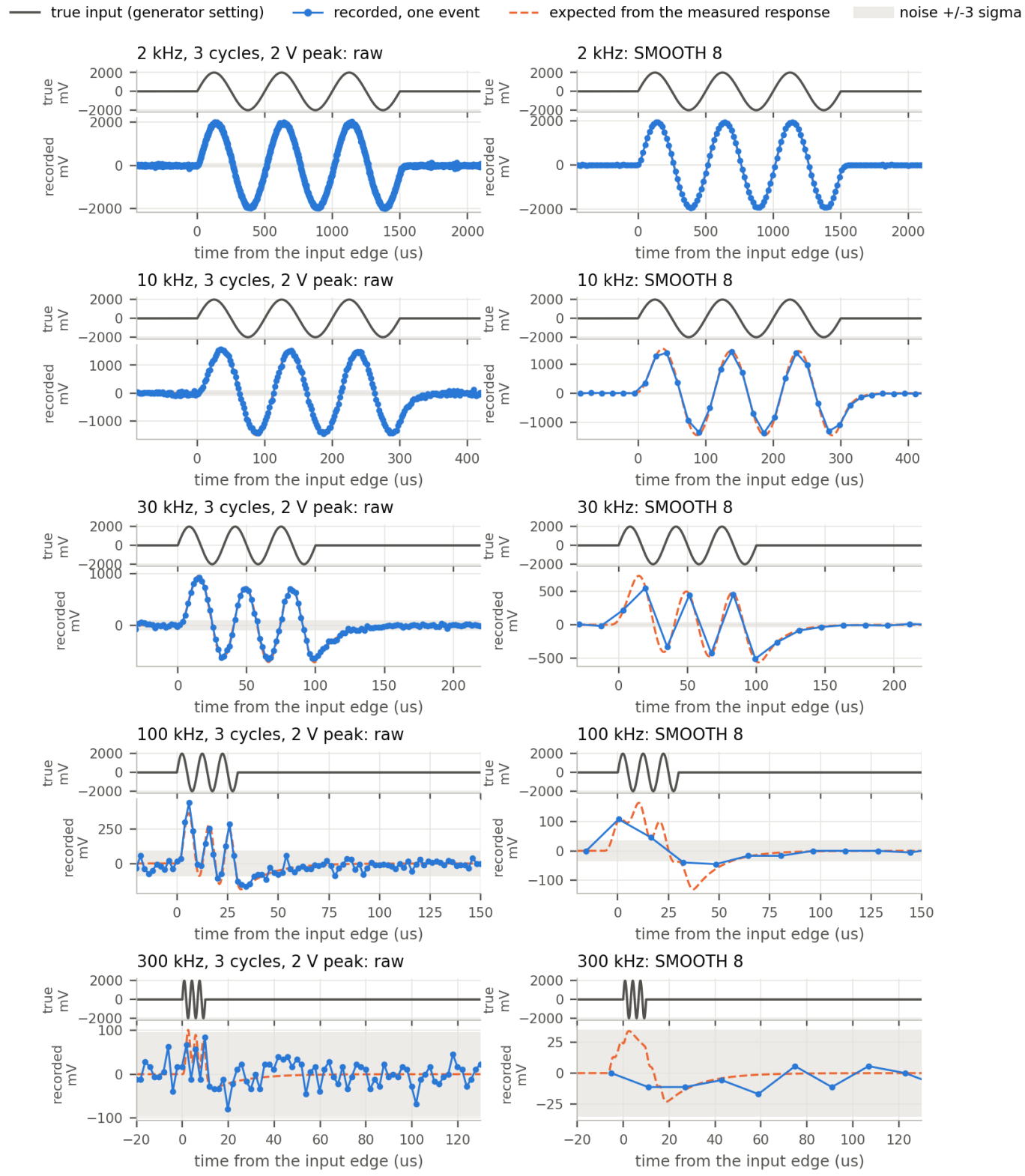


Below: the smallest height that reaches 5 sigma in one shot, for each length, with and without SMOOTH, and with averaging of 16 repeats. The curves flatten above ~50 us (full height reached) and rise as 1/width below ~15 us (the area rule).



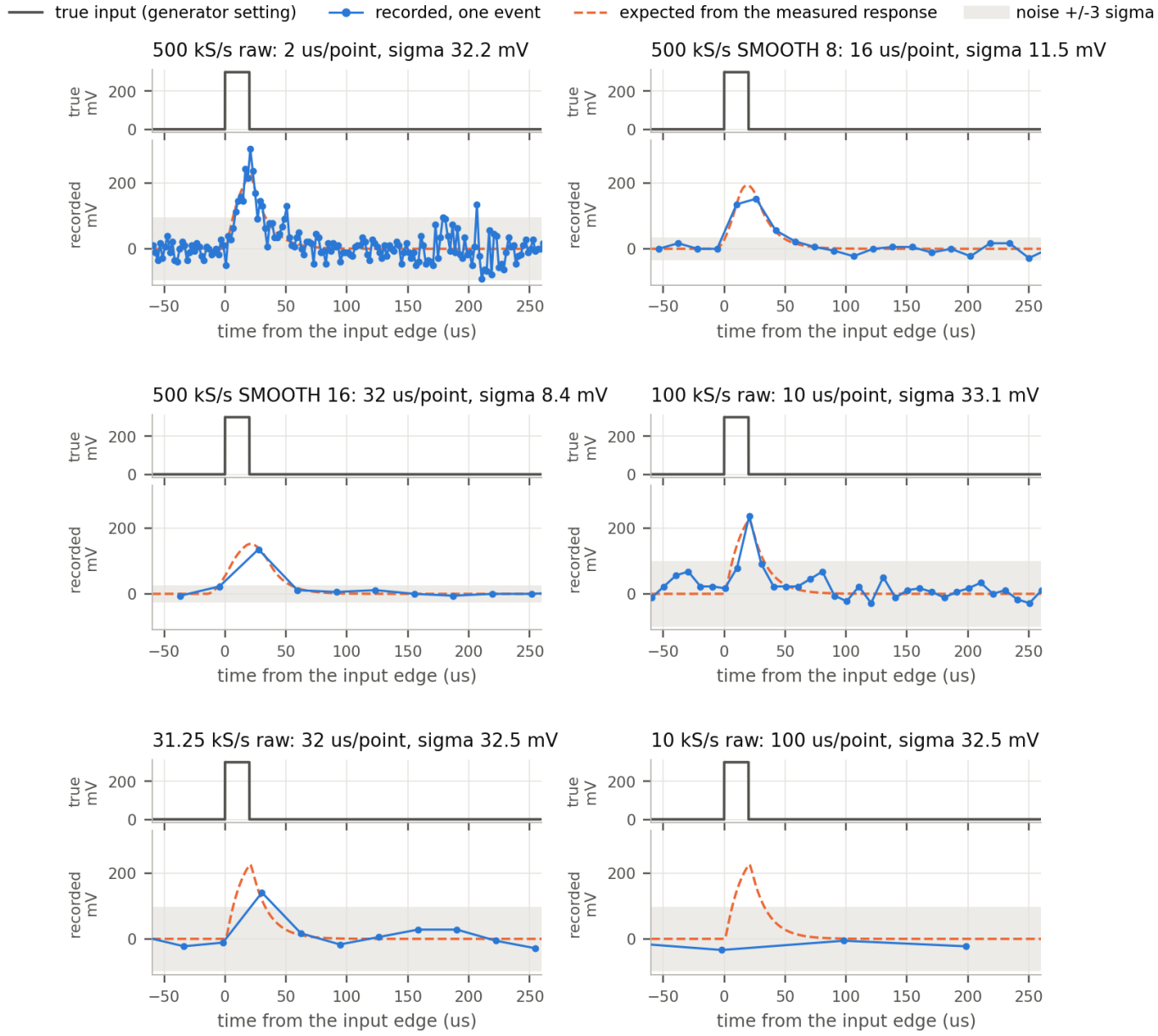
6. Oscillating transients (3-cycle sine bursts, 2 V peak)

Ringing behaves the way the frequency response says. A 2 kHz burst is recorded intact; 10 kHz loses about a quarter and lags by about 12 us; 30 kHz arrives at about a third of its size (the first cycle a little larger); 100 kHz at about a tenth, still clearly visible raw; 300 kHz is a few % and sits at the noise. With SMOOTH 8 (right) the points are 16 us apart: 30 kHz is undersampled and distorted, and 100 kHz merges into one small bump. If the frequency of a ring matters, use raw 500 kS/s.



7. One transient, six settings (20 us, 300 mV)

The board settings decide how much of what the front end passes is kept. At low sample rates without SMOOTH, the points are further apart than the event: at 10 kS/s a point lands on the bump only by luck, and the peak height is usually wrong. SMOOTH at 500 kS/s gives the same point spacing as a lower rate but averages every sample in between, so nothing is stepped over.



for	use	why
weak events, any length	500 kS/s, SMOOTH 8	every sample used, 2.8x lower noise, 16 us/point
the very weakest long events (> 200 us)	500 kS/s, SMOOTH 16-64	more averaging, still faster than the event
the shape of fast edges or ringing	500 kS/s raw	2 us/point, keeps 30-100 kHz content
long records	500 kS/s with SMOOTH raised	not a lower rate: the record keeps 10000 points, so raise SMOOTH (e.g. SMOOTH 50 for 1 s) instead of letting it skip points

8. What this test does not cover

- **A high-impedance source.** The generator drives the input from 50 ohms. With a high-impedance source, the source resistance and the input capacitance form an extra low-pass, and the input picks up mains hum. A 1 Mohm source was measured separately: see Series_1Mohm_source_report.pdf.
- **Signals above +/-10 V.** The generator stops at +/-10 V; +/-12 V is not verified (the pins would sit at 0.61-2.33 V, inside the ADC range).
- **Triggering.** These records are free-running and were lined up afterwards from the generator timing. The board's own trigger fires on one pin's change between consecutive points, and each pin on its own carries the ~632 kHz supply ripple and more noise than the A2-A3 difference. A weak transient that only shows after differencing and SMOOTH may not fire it reliably. Not tested here.
- **Battery power.** All of this was on USB. USB and battery were compared afterwards (2026-11-05) and measured the same noise, ripple and hum, so these results hold on battery too.
- **Polarity and DC level.** Only positive pulses from 0 V were used. The front end is linear across +/-10 V (see the bench run), so negative or offset transients behave the same.

Reproducing this report

py transient_capture.py --board COM32 --label usb (about 7 minutes, drives the Rigol, restores it after)
py transient_report.py runs/<stamp>_transients_usb runs/<stamp>_usb_bench

Data: 20261105-0754_transients_usb and 20261105-0617_usb_bench in Giga_RemoteController_WiFi/frontend_test/runs. Clock offset between the Giga and the Rigol, measured from the data: -110.0 ppm.